Singapore School Kinderland Secondary-2

Computer Science — Term 2 Comprehensive Practice Worksheet

Chapter 1.3 (Storage & Compression) & Chapter 2.2, 2.3 (Transmission & Security)

Date: _____ **Student Name:** _____

Class: Secondary 2 (Sec 2) **Total Marks: 40 Marks | Obtained: _____**

SECTION 1: DATA STORAGE & FILE SIZE CALCULATIONS [10 MARKS]

1. Unit Conversions: (3 Marks)

Convert the following storage capacities into the requested units:

a) Convert 4 KiB into Bytes: _____

b) Convert 2.5 MiB into KiB: _____

c) Explain the difference between 1 Kilobyte (KB) and 1 Kibibyte (KiB):

_____

2. Image File Size Calculation: (3 Marks)

A bitmap image has a resolution of **1024 × 768 pixels** and a colour depth of **16 bits per pixel**.

Calculate the file size in **Kibibytes (KiB)**. Show your full working.

_____

_____

_____

3. Audio File Size Calculation: (4 Marks)

A studio records a **40-second stereo audio clip** with a sampling rate of **44,100 Hz** and a sample resolution of **16 bits**.

Calculate the size of the uncompressed audio file in **Bytes** and in **Mebibytes (MiB)**.

_____

_____

_____

_____

SECTION 2: DATA COMPRESSION & RUN-LENGTH ENCODING [10 MARKS]

4. Lossy vs Lossless Comparison: (4 Marks)

Complete the comparison table by ticking [✓] or stating True/False:

Statement / Scenario	Lossy	Lossless
Decompressed file is 100% identical to the original uncompressed file.		
Permanently removes background sounds that human ears cannot perceive.		
Suitable for executable software program code and medical records.		
Audio format MP3 and image format JPEG.		

5. Run-Length Encoding (RLE) Problem Solving: (6 Marks)

a) A sequence of pixel colours is stored as: `W W W W W B B B B G G G G G G W W`

Write the RLE compressed string using (count, colour) pairs: (3 Marks)

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b) An RLE string is given as: `04 R 03 Y 05 B 02 G`

Decompress the string into its original sequence: (2 Marks)

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c) Under what condition does RLE fail to reduce file size? (1 Mark)

SECTION 3: ERROR DETECTION METHODS [10 MARKS]

6. Parity Block Matrix Error Location: (5 Marks)

A computer transmits data using **Even Parity**. The received parity block grid is shown below:

Byte	Bit 1	Bit 2	Bit 3	Bit 4	Bit 5	Bit 6	Bit 7	Parity Bit
Byte 1	1	0	1	1	0	0	1	0
Byte 2	0	1	1	0	1	1	0	0
Byte 3	1	1	0	1	0	1	1	1
Byte 4	0	0	1	0	1	0	1	1
Parity Byte	0	0	1	1	0	0	1	0

a) Identify which byte contains the transmission error: _____

b) Identify which bit column flipped: _____

c) State the correct value of the corrupted bit: _____

7. Check Digits & ARQ: (5 Marks)

a) Differentiate between a *Transcription Error* and a *Transposition Error*: (2 Marks)

b) In an ARQ protocol, what happens if the sender does not receive an ACK before the timeout timer expires? (3 Marks)

SECTION 4: SYMMETRIC VS ASYMMETRIC ENCRYPTION [10 MARKS]

8. Symmetric Encryption Vulnerability: (3 Marks)

Explain why symmetric encryption suffers from the "Key Distribution Problem" and describe how hackers exploit it:

9. Asymmetric Public/Private Key Encryption Scenario: (4 Marks)

Zayan wishes to send a confidential, encrypted contract to Nadia over the Internet using asymmetric encryption.

a) Which cryptographic key does Zayan use to encrypt the contract?

b) Which key does Nadia use to decrypt and read the contract? _____

c) Can an eavesdropper who intercepts the message decrypt it using Nadia's Public Key? Explain why: (2 Marks)

10. Digital Signatures & SSL/TLS: (3 Marks)

a) What two security properties does a Digital Signature provide? (2 Marks)

b) State the role of a Certificate Authority (CA) in issuing SSL certificates for websites: (1 Mark)